

IoT-Based Machine Fault Detection

Objective: Predict machine failure 24 hours in advance using sensor readings.

Datasets: NASA C-MAPSS (FD001, FD002, FD003, FD004)

Best Model: Logistic Regression | ROC-AUC: 0.617

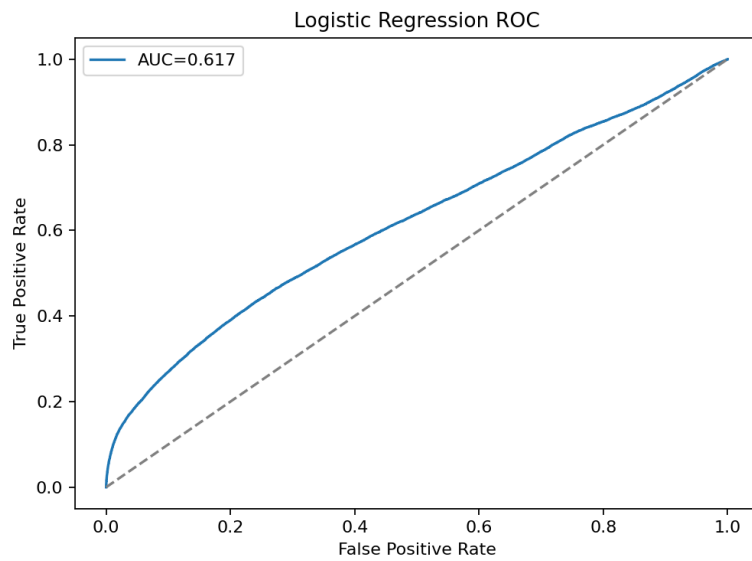
Threshold: 0.003

Methods

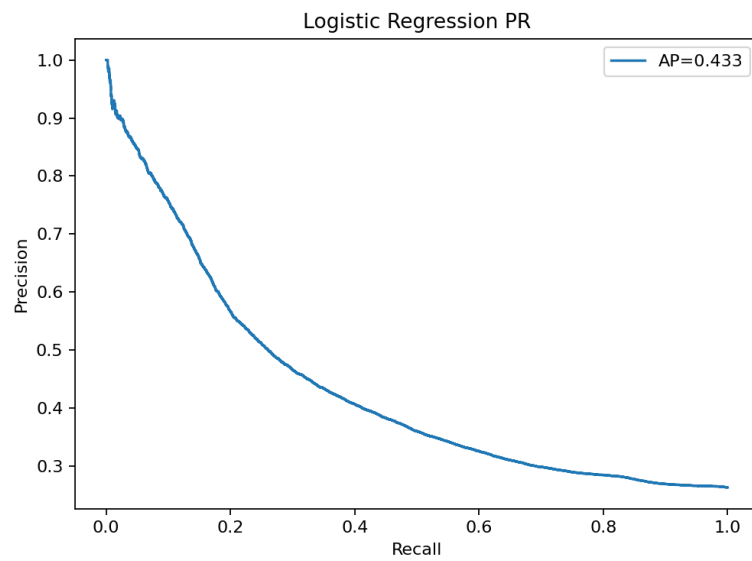
- Preprocessing: anomaly clipping (IQR), per-FD label generation (RUL→binary ≤ 24), per-train scaling (Min-Max).
- Feature Engineering: rolling mean/std (window=5), rate-of-change and smoothed ROC, op-setting interaction.
- Models: Logistic Regression, Decision Tree with GridSearchCV (5-fold Stratified).
- Metrics: ROC-AUC, PR-AUC, Confusion Matrix; threshold chosen to maximize F1 (or set precision floor).

Results

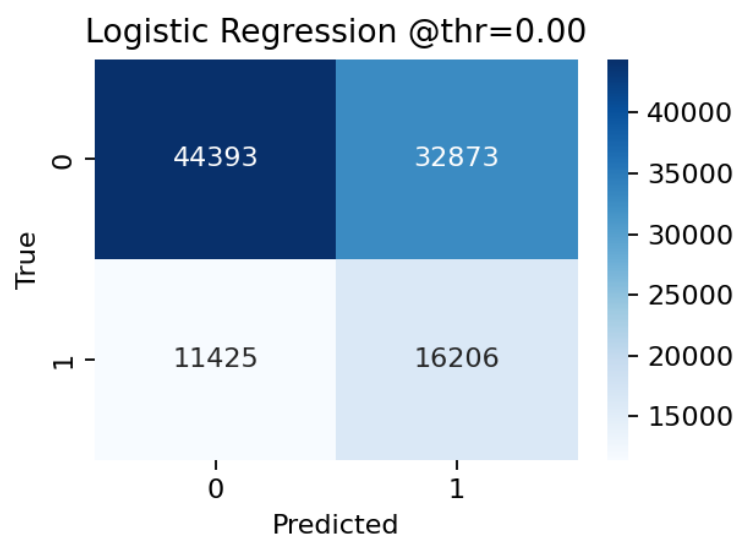
ROC Curve



Precision-Recall Curve



Confusion Matrix



Business Case

Assumptions: Cost of failure = \$10,000, Preventive maintenance = \$1,000.

Baseline (do nothing) cost: \$276,310,000

Strategy total cost: \$163,329,000

Estimated savings: \$112,981,000

Conclusion

- The selected model (Logistic Regression) achieves ROC-AUC 0.617 with an operational threshold of 0.00
- Rolling and interaction features capture degradation patterns; dashboard supports per-machine risk tracking
- Next steps: add LSTM/GRU sequence model baseline, calibrate probabilities (Platt/Isotonic), and deploy th